Acute Kidney Injury and Kidney Replacement Therapy in the Critically Ill Patient

Andrea M. Nei, Pharm.D., FCCM, BCPS, BCCCP

Mayo Clinic

Rochester, Minnesota

Acute Kidney Injury and Kidney Replacement Therapy in the Critically Ill Patient

Andrea M. Nei, Pharm.D., FCCM, BCPS, BCCCP

Mayo Clinic

Rochester, Minnesota

Learning Objectives

1. Define acute kidney injury (AKI).
2. Differentiate between common categories of drug-induced kidney disease.
3. Discuss key principles of continuous kidney replacement therapy (KRT), including indications, timing, and circuit components.
4. Apply drug-dosing concepts in continuous KRT to estimate a sieving coefficient, saturation coefficient, and/or drug clearance on the basis of drug characteristics and device settings.

Abbreviations in This Chapter

AIN	Acute interstitial nephritis
AKI	Acute kidney injury
AKIN	Acute Kidney Injury Network
ARF	Acute renal failure
ATN	Acute tubular necrosis
CKD	Chronic kidney disease
CVVH	Continuous venovenous hemofiltration
CVVHD	Continuous venovenous hemodialysis
CVVHDF	Continuous venovenous hemodiafiltration
DIKD	Drug-induced kidney disease
EDD	Extended daily dialysis
GFR	Glomerular filtration rate
ICU	Intensive care unit
IHD	Intermittent hemodialysis
KDIGO	Kidney Disease: Improving Global Outcomes
KRT	Kidney replacement therapy
MW	Molecular weight
RIFLE	Risk, injury, failure, loss, end-stage renal disease
SCr	Serum creatinine
SLED	Sustained low-efficiency dialysis
UOP	Urinary output
Vd	Volume of distribution

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

Questions 1–3 pertain to the following case.

E.R. is a 67-year-old man admitted to your intensive care unit (ICU) several days ago with acute respiratory failure. He required mechanical ventilation and was placed on empiric antibiotics to cover community-acquired pneumonia. His kidney function has worsened over the past 5 days (serum creatinine [SCr] 0.6 mg/dL on admission; today, 2.4 mg/dL), and he now requires kidney replacement therapy (KRT) with continuous venovenous hemofiltration (CVVH). He is extubated, but still unstable. His current medications include as-needed fentanyl, norepinephrine, ceftriaxone, azithromycin, and heparin prophylaxis.

1. In E.R., which drug-induced kidney disease phenotype is most associated with his antibiotic program?
   A. AKI.
   B. Glomerular disease.
   C. Nephrolithiasis/crystalluria.
   D. Tubular dysfunction.

2. What KDIGO stage best characterizes E.R.'s AKI?
   A. Stage 1.
   B. Stage 2.
   C. Stage 3.
   D. Cannot determine because urinary output (UOP) is not provided.

3. E.R. continues to worsen. He is now febrile with an increasing white blood cell count (WBC). You are asked to dose his antibiotics while he receives CVVH. Which would be the best place to begin looking for dosing recommendations?
   A. Intermittent hemodialysis (IHD) guidelines.
   B. Package insert.
   C. Primary literature/dosing summaries.
   D. Estimates using an estimated sieving coefficient.

4. Which best describes the method for solute removal during CVVH?
   A. Convection.
   B. Diffusion.
   C. Both convection and diffusion.
   D. Membrane binding.

5. Which drug would most likely be cleared through a CVVHD circuit?
   A. Drug A: 10% protein bound, 140-Da molecular weight (MW), volume of distribution (Vd) 1.2 L/kg.
   B. Drug B: 90% protein bound, 1030-Da MW, Vd 0.8 L/kg.
   C. Drug C: 40% protein bound, 250-Da MW, Vd 1.5 L/kg.
   D. Drug D: 35% protein bound, 480-Da MW, Vd 4.6 L/kg.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   Task A: 1, 2, 4
   Task B: 4
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1, 2, 5, 8
   b. Task B: 3
   c. Task C: 3, 4
3. Domain 3: Professional Practice
   a. Task B: 1, 2, 4, 6

I. ACUTE KIDNEY INJURY

A. Introduction
1. Acute kidney injury (AKI), previously known as acute renal failure, is an acute decline in kidney function and encompasses the full continuum of kidney injury and functional impairment.
2. AKI is a decrease in kidney function that occurs over hours to days.
3. AKI results in the accumulation of waste products and, as urine volume decreases, metabolic disturbances and fluid retention.
4. AKI has been associated with increased mortality, development of chronic kidney disease (CKD), cardiovascular complications, and end-stage renal disease.
5. AKI is a heterogeneous syndrome with multiple causes and associated pathophysiologic mechanisms.

B. Epidemiology
1. The incidence of AKI varies and depends on the definition used, its cause, and the patient population (e.g., community- or hospital-acquired and severity of illness).
2. Hospital-acquired AKI is more common in high-income countries, with the highest prevalence in intensive care units, whereas community-acquired AKI is more common in lower-income settings.
3. In low- to middle-income countries community-acquired AKI tends to be caused by a single disease, including dehydration, sepsis, toxins (bites), and pregnancy. Other causes include HIV infection, hantavirus infection, and malaria or dengue disease. Patients tend to be younger, previously healthy, and have more difficult access to care.
4. Hospital-acquired AKI occurs infrequently in patients with less severe illness admitted to a general hospital ward (1.9%–20%). In critically ill patients, the risk is greater. In this group, AKI is estimated to occur in 20%–67% of patients.
5. Sepsis and shock are common causes of acute tubular necrosis (ATN), which is a leading cause of AKI in critical illness. Other risk factors for AKI include use of intravenous radiocontrast agents, major surgery (especially cardiothoracic), nephrotoxic medications, and chronic medical conditions (e.g., history of CKD, congestive heart failure, and diabetes). Most patients have more than one risk factor.
6. Mortality rates in patients with AKI are 10%–80%, with the highest in patients with multisystem organ failure (50%) and those requiring kidney replacement therapy (KRT) (up to 80%).
7. Women tend to be underrepresented, with 60% of patients with AKI being male, which may be related to limited access to healthcare in women or sex-related risks for AKI development in males. Ethnic and racial differences are less well described. A cohort study in the United States found socioeconomic status to account for higher risk of AKI in African American individuals compared with white individuals.

C. Definitions (Table 1)
1. During the past several decades, many definitions have been used for AKI, making it difficult to compare patient populations across studies. In 2004, the Acute Dialysis Quality Initiative workgroup developed the RIFLE (risk, injury, failure, loss, end-stage renal disease) definition and staging system.
   a. RIFLE categorizes AKI into three grades of increasing severity (risk, injury, and failure) and two clinical outcomes (loss and end-stage).
   b. Staging is based on the degree of SCr increase or a decrease in glomerular filtration rate over 7 days, or the duration of oliguria or anuria.
2. Because of emerging data suggesting that even small changes in kidney function lead to worse outcomes, the Acute Kidney Injury Network (AKIN) criteria was developed.
   a. Similar to RIFLE, this group defined AKI using a staging system of 1-3, defined by a reduction in kidney function over no more than 48 hours using measures of SCr, urinary output (UOP), and need for KRT.

b. The primary difference between the RIFLE and the AKIN criteria is that AKIN includes a small absolute change in SCr (0.3 mg/dL or more) as part of the diagnostic criteria for AKI and sets a 48-hour time constraint for the criteria to be met. AKIN also omits the outcome classifications (loss and end-stage) previously proposed in the RIFLE criteria.
c. In patients requiring KRT, AKIN stage 3 is met, regardless of the stage they are in when KRT is initiated.
d. Several studies have validated these criteria and show that the more severe the RIFLE class or AKIN stage, the worse the clinical outcome.

3. In 2012, a third consensus definition was introduced, the Kidney Disease: Improving Global Outcomes (KDIGO) classification system, which is the current criterion standard.
   a. Separately, the RIFLE criteria and the AKIN criteria were each shown to be suboptimally sensitive for detecting AKI.
   b. The KDIGO criteria combine the strengths of both RIFLE and AKIN, retain the AKIN criteria of a rise in SCr of 0.3 mg/dL within 48 hours, and allow 7 days for a 50% increase in SCr from baseline, as seen in RIFLE.
4. Despite these updates, limitations exist with current definitions.
   a. Current diagnostic markers (SCr and urine output) reflect loss of function rather than damage or injury.
   b. SCr is an imperfect biomarker for estimation of glomerular filtration rate and is significantly affected by muscle mass. In the critically ill, production may be decreased and volume of distribution may be affected by volume overload, both decreasing SCr levels.
   c. Baseline SCr values may not be available in all patients. Various techniques may be used such as admission SCr, inpatient nadir SCr, imputed creatinine using eGFR 75 mL/min/1.73m2, or preadmission baseline, all with certain strengths and limitations.
   d. Decreased urine output is a physiologic response to intravascular volume depletion and may not imply tubule injury. It is also confounded by use of diuretics.
   e. Current consensus definitions do not address subphenotypes of AKI, which may support development of therapeutics or trial enrichment for management of AKI.

Table 1. Criteria for Establishing AKI

RIFLE		AKIN		KDIGO		All Three Classification Systems
Stage	**SCr/GFR Criteria**	**Stage**	**SCr Criteria**	**Stage**	**SCr Criteria**	**UOP Criteria**
R	Increase to ≥ 1.5-fold or GFR decrease > 25% from baseline	**1**	Increase to 1.5- to 2-fold above baseline or by ≥ 0.3 mg/dL within 48 hr	**1**	Increase to 1.5- to < 2-fold above baseline over 7 days or by ≥ 0.3 mg/dL within 48 hr	< 0.5 mL/kg/hr for 6-12 hr
I	Increase to ≥ 2-fold or GFR decrease > 50% from baseline	**2**	Increase to ≥ 2- to 3-fold above baseline	**2**	Increase to ≥ 2- to < 3-fold above baseline	< 0.5 mL/kg/hr for ≥ 12 hr
F	Increase to ≥ 3-fold, GFR decrease > 75% from baseline, or SCr ≥ 4 mg/dL (acute increase of at least 0.5 mg/dL)	**3**	Increase > 3-fold above baseline or ≥ 4 mg/dL with an acute rise of ≥ 0.5 mg/dL or on KRT	**3**	Increase ≥ 3-fold above baseline or ≥ 4 mg/dL or initiation of KRT	< 0.3 mL/kg/hr for ≥ 24 hr or anuria for ≥ 12 hr

Table 1. Criteria for Establishing AKI (*continued*)

RIFLE		AKIN		KDIGO		All Three Classification Systems
Stage	**SCr/GFR Criteria**	**Stage**	**SCr Criteria**	**Stage**	**SCr Criteria**	**UOP Criteria**
L	Complete loss of function for > 4 wk					
E	Complete loss of function for > 3 mo					

AKI = acute kidney injury; AKIN = Acute Kidney Injury Network; GFR = glomerular filtration rate; KDIGO = Kidney Disease: Improving Global Outcomes; KRT = kidney replacement therapy; RIFLE = risk (R), injury (I), failure (F), loss (L), end-stage kidney disease (E); SCr = serum creatinine; UOP = urinary output.

5. Kidney dysfunction/damage is a spectrum.
   a. In 2012, KDIGO proposed four categories to encompass the spectrum of kidney function (Figure 1; Table 2 provides detailed definitions).
      i. No known kidney disease (NKD)
      ii. AKI: Acute kidney dysfunction over 7 days or less
      iii. Acute kidney disease (AKD): For those who meet the AKI definition or with other types of kidney dysfunction/damage for less than 3 months
      iv. Chronic kidney disease (CKD): Kidney dysfunction/damage for more than 3 months
   b. The definitions were intended to improve consistency in applied nomenclature in care, research, and public health and to identify patients with AKD who may need therapy to restore kidney function and mitigate or reverse kidney damage.
   c. The definition of AKI recovery is not well defined; however, identification of early resolution within days versus delayed resolution of weeks to months, has important prognostic implications. Nonresolving AKI is associated with greater risk of major adverse kidney events, specifically the development of chronic kidney disease.
   d. Increasingly, patients in the AKD subgroup are targeted for interventions designed to prevent CKD development, including bundled strategies delivered in AKI follow-up clinics. Interprofessional survivor clinics for AKI are reported to improve patient education, engagement, follow-up, and outcomes. Pharmacists can play a central role in such initiatives by providing a careful medication evaluation (renally eliminated and nephrotoxic drugs), monitoring and supportive care for key concurrent conditions (e.g., hypertension, lipid management), and patient and caregiver education.

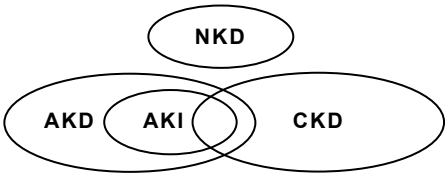


Figure 1. Overview of the spectrum of kidney disease.

AKD = acute kidney disease; AKI = acute kidney disease; CKD =chronic kidney disease; NKD = no known kidney disease.

Table 2. Definitions of NKD, AKI, AKD, and CKD According to Function and Structure

	Functional Criteria (change in SCr, GFR,[a] or UOP)	**Structural Criteria (damage[b] or no damage and duration)**
NKD	GFR ≥ 60 mL/min/1.73 m^2, stable SCr	No
AKI	Increase in SCr by 50% within 7 days Or Increase in SCr by 0.3 mg/dL within 2 days Or Oliguria	No criteria
AKD	AKI Or GFR < 60 mL/min/1.73 m^2 for < 3 mo Or Decrease in GFR by ≥ 35% or increase in SCr by > 50% for < 3 mo	Kidney damage for < 3 mo
CKD	GFR < 60 mL/min/1.73 m^2 for > 3 mo	Kidney damage for > 3 mo

[a]Assessed from measured or estimated GFR.

[b]Kidney damage is assessed by pathology, biomarkers (urine or blood), imaging, and kidney transplantation (for CKD).

AKD = acute kidney disease and disorders; CKD = chronic kidney disease.

D. Diagnostic Workup and Evaluation
   1. There is no single specific test for diagnosing AKI.
   2. A thorough history and complete examination should be completed, including rate of kidney function decline, symptoms, and coexisting diseases
   3. A comprehensive review of current and recent medications should be conducted. Drug-related injury is a leading cause of AKI.
   4. Chemistries (blood urea nitrogen [BUN], SCr, serum electrolytes, albumin, and a complete blood cell count) and a urinary analysis (microscopy, sodium, and osmolality) may assist in determining the type of failure (e.g., prerenal, intrinsic, postrenal).
   5. If a bladder catheter is not present, it is reasonable to insert one in most cases. Compelling indications for catheterization include cases in which ins and outs are difficult to measure accurately such as in patients with incontinence or those with evident postvoid residuals. If a catheter is present without UOP, it should be evaluated for patency.
   6. Abdominal compartment syndrome should be ruled out if clinically suspected. Acute oliguria and AKI can be the result of increasing renal outflow pressure and reduced kidney perfusion.
   7. Routine use of kidney ultrasonography is of limited usefulness because most ICU-related AKI is associated with decreased effective arterial blood volume and/or ATN. However, it may be useful in high-risk patients or after an initial evaluation fails to reveal the cause of AKI.
   8. Renal biopsies have limited usefulness but may be necessary. They are most useful in intrinsic renal failure not associated with ATN or when glomerulopathy is suspected.
   9. The "furosemide stress test" has been used as a tool to predict AKI progression. Tubular dysfunction after AKI may affect diuretic response from loss of sodium transporter expression, Na-K-ATPase redistribution and organic acid transporter mistargeting. A dose of 1 mg/kg in patients who are diuretic naive and 1.5 mg/kg in patients who are non-diuretic naive is typically used. Urinary output cutoff points vary across studies, but is most commonly defined as 200 mL over 2 hours.

10. Biomarkers: Recently, significant advances were made in the field of biomarkers for predicting and detecting AKI (Figure 2). When the KDIGO guidelines were released in 2012, these data were not available; thus, they were not included in the criteria for the diagnosis of AKI. Despite emerging literature and calls for revision of AKI classification to incorporate biomarkers, the most recent 2019 KDIGO controversies conference on AKI suggests that additional research is needed before including these biomarkers in the diagnostic criteria for AKI.

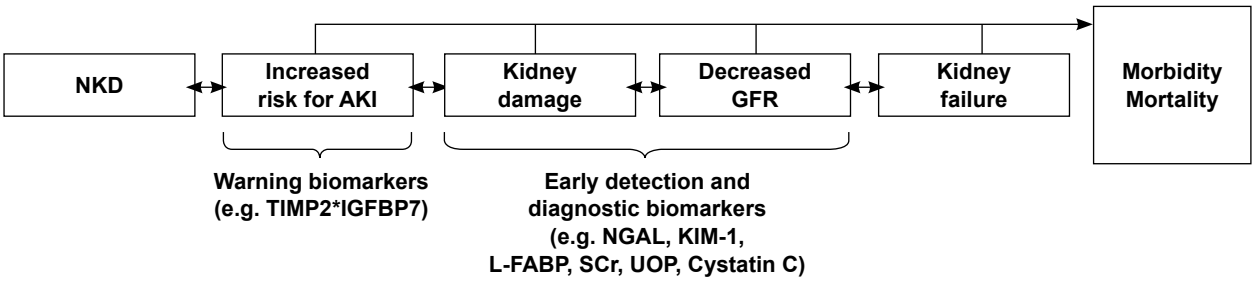


Figure 2. Evolution of AKI as it corresponds to biomarker release.
Adapted from: Bellomo R, Kellum JA, Ronco C. Acute kidney injury. Lancet 2012;380:756-66.

a. Functional markers: Indicate the kidney's capacity for clearance and are used to estimate GFR and functional loss; these can help indicate a need for drug dosing changes
   i. Examples: SCr, UOP, serum cystatin C, proenkephalin, beta-trace protein, β2 microglobulin
   ii. These biomarkers may rise with or without evidence of structural damage to the kidney (e.g., prerenal AKI, which could correspond to a reduced GFR that may not yet have resulted in kidney damage).
   iii. Recent evidence suggests that cystatin C can predict the pharmacokinetics of medications as well as, if not better than, SCr with certain renally eliminated medications. Drug-specific dosing models need to be developed and tested.

b. Damage/injury markers: Signal injury to the kidney cells or at least cellular distress; these can help signal patients who would benefit from limiting exposure to nephrotoxins
   i. Warning biomarkers (e.g., cell-cycle arrest markers [TIMP-2•IGFBP7])
      (a) TIMP-2•IGFBP7 is a U.S. Food and Drug Administration–approved urinary test; these biomarkers are released during a pause (an "arrest") in mitosis when the biomarker indicates the cell may be injured and should not divide.
      (b) In the PrevAKI (Intensive Care Med 2017;43:1551-61) and BigpAK (Ann Surg 2018;267:1013-20) trials, surgical patients with a high risk of AKI on the basis of [TIMP-2•IGFBP7] greater than 0.3 who were randomized to an AKI prevention bundle had a lower incidence of AKI than did individuals who received usual care. This is a significant advancement because it suggests that even in high-risk patients, AKI can indeed be prevented.
      (c) A retrospective cohort study of the ProCESS trial by the ProGReSS-AKI investigators (JAMA Netw Open 2022:5:e2212709) evaluated the incorporation of TIMP-2•IGFBP7 assessment (TIMP-2•IGFBP7 greater than 2.0) into the classification of AKI as well as the association with survival among patients with septic shock and similar AKI stage classification. In patients with existing AKI, a positive biomarker was associated with lower survival and higher mortality at 30 days. These data suggest that incorporation of cell-cycle arrest biomarkers may augment AKI staging in patients who meet functional criteria for AKI.

ii. Early detection and diagnostic biomarkers (e.g., NGAL, KIM-1, L-FABP, NAG, urine cystatin C, CCL14) are another set of tools that have been studied as early indicators of kidney damage or predictors of persistent AKI; each test has different performance characteristics

iii. The cost-effectiveness of using these biomarkers to inform routine care has not been well studied; however, given the high costs of AKI, particularly when dialysis is required, a systematic approach to biomarker use could prove economical. Additional research is needed to explore this further.

c. Although the evidence is promising, novel biomarker use is not yet mainstream. Several factors, including the evidence base that is still in its infancy, implementation challenges, and cost, have precluded widespread adoption as an adjunct to SCr and UOP at this time. As use continues, additional information on the logistics of successfully applying these tests is expected.

11. Kinetic eGFR is another strategy that has been proposed to evaluate rapidly changing kidney function (J Am Soc Nephrol 2013;24:877-88). Creatinine is a lagging marker of kidney function and may take 48 hours to increase from the onset of kidney damage. The kinetic eGFR equation is thought to provide a quantitative solution to estimate kidney function in this difficult scenario. Using a simple algebraic formula, including the initial creatinine content, the volume of distribution, the creatinine production rate, and the change in SCr over time, inputs can be used to quantitatively interpret creatinine change. Kinetic eGFR predicted AKI and the need for KRT more accurately than Modification of Diet in Renal Disease (Clin Kidney J 2017;10:202-8). Published literature on the use of kinetic eGFR to inform medication dosing remains in infancy.

E. Causes and Mechanisms of AKI

1. Prerenal: AKI attributable to decreased kidney perfusion. Decreased kidney perfusion could be caused by low effective arterial blood volume, as in hypovolemia or acute decompensated heart failure, peripheral vasodilation (as with sepsis), or renal vasoconstriction or renal occlusion (caused by hepatorenal syndrome or arterial thrombosis, respectively). Some have suggested to do away with the nomenclature "prerenal" because it could be misinterpreted to mean hypovolemia, which only reflects a small subset of cases with decreased kidney perfusion.
2. Renal (intrinsic): AKI secondary to parenchymal or vascular diseases. Acute tubular necrosis (ATN) is the most commonly cited example of this category of AKI in the hospital setting. AKI secondary to vasculitis, glomerulonephritis, interstitial nephritis, infections, connective tissue disease, and crystalluria are in this category as well.
3. Postrenal: AKI associated with urinary tract obstruction. Common causes include prostatic hypertrophy, nephrolithiasis, and tumor obstruction. If an obstruction is suspected, placing a urinary catheter may be a sufficient temporizing measure. If higher in the urinary tract, nephrostomy tubes may be warranted to alleviate hydronephrosis.

F. Drug-Induced Kidney Disease (DIKD)

1. Drugs are associated with about one-third of AKI cases in the hospital. The concept of "nephrotoxin stewardship" has been proposed. *Nephrotoxin stewardship* is a term taken from antimicrobial stewardship that generally refers to "coordinated interventions to improve and measure the appropriate use of drugs by promoting the selection of the optimal drug regimen, including dosing, duration of therapy, and route of administration" (Clin Infect Dis 2016;62:e51-77). Kane-Gill has proposed three goals for nephrotoxin stewardship: coordinated strategies to enhance medication safety, coordinated strategies to ensure kidney health, and avoidance of unnecessary costs (Crit Care Clin 2021;37:303-20).
2. A standardized classification system (Kidney Int 2015;88:226-34) was proposed for DIKD that classifies kidney dysfunction according to three primary factors:

   a. Phenotype of kidney injury: AKI (ATN, acute interstitial nephritis [AIN], osmotic nephrosis), glomerular disorder (hematuria, proteinuria), nephrolithiasis (crystalluria, nephrolithasis, obstruction), and tubular dysfunction (renal tubular acidosis, Fanconi syndrome, syndrome of inappropriate diuretic hormone, diabetes insipidus, phosphate wasting)
   b. Mechanism: Type A (dose-dependent toxicities), type B (unpredictable, not dose-dependent)
   c. Time course: Acute (1–7 days), subacute (8–90 days), chronic (more than 90 days)
3. More recently, a group of experts proposed a more contemporary DIKD schema that incorporates functional changes and tissue damage biomarkers (Crit Care 2023;27:435). The four categories include: "dysfunction without damage", "damage without dysfunction", "both dysfunction and damage," and "neither dysfunction nor damage".
4. Extrarenal: Drugs that alter renal perfusion or influence intraglomerular hemodynamics can heighten a patient's risk of AKI. Examples include the following:
   a. Loop diuretics: Intravascular volume depletion leading to loss of effective arterial blood volume
   b. Negative inotropes or vasodilatory antihypertensives that reduce renal perfusion
   c. Drugs that alter intraglomerular hemodynamics: Nonsteroidal anti-inflammatory drugs (NSAIDs), angiotensin-converting enzyme inhibitors/angiotensin receptor blockers (ACEIs/ARBs)
      i. Normally, glomerular pressure is maintained, in part, by vasodilation of the afferent arterioles or vasoconstriction of the efferent arterioles.
      ii. Vasodilation of the afferent arteriole is partly controlled by the effects of prostaglandins. Medications that decrease prostaglandin synthesis such as NSAIDs decrease the ability of the afferent arteriole to vasodilate and thus increase the risk.
      iii. Vasoconstriction of the efferent arteriole, which helps maintain transcapillary pressure, is mediated through angiotensin II. Drugs that block angiotensin II such as ACEIs and ARBs prevent effective efferent vasoconstriction and thus increase the risk.
      iv. The so-called "triple whammy" of NSAIDs/diuretics/ACEIs or ARBs has been described as having a uniquely high risk of AKI because of the multiplicity of the intraglomerular hemodynamic impact (BMJ 2013;346:e8525). Information from an adverse drug event report database showed that the median time to AKI onset was 20 days with a single triple whammy drug, 44 days for two drugs, and 8 days for three drugs. The onset of AKI is generally early and decreases over time, especially if an NSAID is part of the combination therapy (Plos One 2022;17:e0263682).
      v. Acute use of calcineurin inhibitors, particularly in the setting of other risk factors for AKI, is thought to interfere with renal hemodynamics.
5. Intrarenal toxicity: There are several types of kidney toxicity from xenobiotics. These include direct tubular toxicity, AIN, glomerular diseases, osmotic nephrosis, crystalluria/nephrolithiasis, and altered electrolyte handling.
   a. Direct tubular toxicity: Drugs can be associated with direct tubular damage, cell apoptosis, and necrosis. ATN may be suggested by the presence of granular casts ("muddy brown casts") or renal tubular epithelial cells on a urinalysis.
      i. Examples of drugs that contribute to ATN: Intravenous contrast, aminoglycosides, amphotericin B, vancomycin, colistin, antiviral agents, platinum chemotherapy
      ii. Direct tubular toxicity is often a type A injury that responds to dose adjustment or drug discontinuation and supportive care.

b. AIN: Drug-associated AIN is thought to be a cell-mediated immune response, analogous to a type 4 hypersensitivity reaction. Microscopic hematuria, sterile pyuria, and nonnephrotic range proteinuria are features of AIN on urinalysis. Other features of AIN can include urine and/or peripheral eosinophilia, clinical symptoms of fever, rash, and joint pains. Although commonly cited in the literature, these may not be universally present; in fact, only 10%–30% of patients with this diagnosis have the "classic triad" of eosinophilia, fever, and rash. A kidney biopsy can be considered but, in the acute setting, is rarely performed until other causes are ruled out.
   i. Drug causes have been implicated in up to 75% of AIN cases. Other separate or concurrent contributors to AIN include infections (5%–10%), idiopathic causes (5%–10%), and causes associated with systemic diseases (10%–15%).
   ii. Examples of drugs that contribute to AIN: Penicillins, cephalosporins, sulfonamides, ciprofloxacin, vancomycin, NSAIDs and cyclooxygenase-2 [COX-2] inhibitors, proton pump inhibitors, immune checkpoint inhibitors, phenytoin, valproic acid, ranitidine, diuretics, and cocaine
   iii. Management of AIN includes discontinuation of the offending agent. These injuries are often subacute and take weeks to months to resolve. Steroids may be used, but this remains controversial.
   iv. Recently, the combination of piperacillin/tazobactam and vancomycin received specific attention for its risk of nephrotoxicity. In ICU patients, observational studies suggest that empiric courses of less than 72 hours are no more nephrotoxic than other such combinations. Preclinical models in rats using histopathology and novel urinary kidney biomarkers suggest no evidence of heightened risk with the combination (J Antimicrob Chemother 2020;75:1228-36). In addition, a prospective cohort study (Intensive Care Med 2022;48:1144-55) identified an increased risk of creatinine-defined AKI with piperacillin/tazobactam and vancomycin combination compared with the cefepime and vancomycin combination, but not changes in cystatin C, blood urea nitrogen, mortality, or need for KRT. This finding supports the idea that the piperacillin/tazobactam and vancomycin effects on creatinine may represent a pseudo-toxicity rather than true AKI. In observational data in humans, evidence still favors a potential increased risk of AKI with the combination and remains an ongoing area of research; thus, the risk-benefit of other combinations should be considered in conjunction with other potential toxicities or antibiotic stewardship efforts.

c. Glomerular diseases: DIKD rarely manifests as glomerular disease. Glomerular diseases from medication can include immune-mediated glomerular diseases, including antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis with necrotizing and crescentic/pauci-immune glomerulonephritis, drug-induced lupus, and drug-associated membranous nephropathy. In addition, drugs may cause direct glomerular toxicity, resulting in minimal change disease or focal segmental glomerulosclerosis. Depending on the type of injury, suggestive diagnostics of drug-associated glomerular diseases may include proteinuria, hematuria, positive antibody tests, rash, fever, myalgias, and polyarthritis. A kidney biopsy is often pursued.
   i. Examples of drugs that contribute to immune-mediated glomerular diseases include hydralazine and propylthiouracil. Drugs that contribute to direct glomerular cell toxicity include lithium, NSAIDs, sirolimus, anabolic steroids, anti-angiogenesis drugs (i.e., bevacizumab, sunitinib, sorafenib), chemotherapeutics (i.e., mitomycin C, gemcitabine), calcineurin inhibitors, interferon, thienopyridines (i.e., clopidogrel, prasugrel), and quinine.
   ii. Management of drug-induced glomerular disease includes identifying the subtype and discontinuing the offending agent. These injuries are often type B reactions, subacute or chronic, and may partly be irreversible. Immunosuppressants and plasmapheresis may be indicated for treatment.

d. Osmotic nephrosis: Drugs such as mannitol; hydroxyethylstarch; immunoglobulins, especially those containing sucrose; and dextrans may lead to osmotic nephrosis. In these cases, drug molecules undergo pinocytosis and cellular accumulation, and the cells become swollen and vacuolated, leading to tubular lumen obstruction.
e. Crystalluria/nephrolithiasis: Certain medications are insoluble in the urine and can result in crystalline nephropathy and/or kidney stones. Suggestive diagnostics include crystalluria on urinary analysis, overt nephrolithiasis, and ultrasound findings indicative of obstruction. Examples of drug culprits include antibiotics, acyclovir, methotrexate, antiretrovirals, triamterene, and ethylene glycol–associated calcium oxalate crystal precipitation.
f. Altered electrolyte handling/tubular dysfunction: Drugs may alter how the kidney handles electrolytes, tubular proteins, and water. Example patterns include the syndrome of inappropriate antidiuretic hormone, an acquired Fanconi syndrome; renal tubular acidosis; and nephrogenic diabetes insipidus. Serum and urine electrolyte abnormalities, altered serum and urine osmolality, and an altered acid-base profile may each indicate this type of DIKD. Examples of drug causes include antiretrovirals, lithium, selective serotonin reuptake inhibitors, antiepileptics, and vincristine.

G. AKI Survivorship
1. Recently, increased emphasis has been placed on AKI survivor care. AKI survivor care focuses on patients with an episode of AKI during hospitalization who are being discharged. Individuals have tried specific AKI survivor clinics to improve comprehensive kidney care follow-up to limit the burden of CKD.
2. The landmark FUSION randomized controlled trial was recently published (Clin J Am Soc Nephrol 2021;16:1005-14). Patients who survived an episode of stage 2 or 3 AKI were randomized to either early nephrology follow-up with a bundled care intervention or usual care. Recruitment was slow, and although process outcomes were improved (more nephrology visits, more kidney function monitoring, more medication reconciliation and review) clinical outcomes, including the primary outcome, which was a major acute kidney event at 1 year, were no different. More research is needed to determine the optimal method for follow-up in AKI survivors that is aligned with best practices and patient-centeredness.

Patient Case

Questions 1–3 pertain to the following case.
F.B. is a 68-year-old male patient (weight 70 kg) admitted to your ICU with fever, elevated WBC, respiratory failure requiring mechanical ventilation, and norepinephrine to support his blood pressure. His medical history is significant for chronic back pain, diabetes, and hypertension. He takes both enalapril and glipizide once daily, as well as acetaminophen as needed. Before this admission, he had otherwise been healthy. He saw his primary care provider about 1 week ago. At that time, his blood pressure was 140/80 mm Hg, and his hemoglobin A1C was 5.2%. His laboratory workup was also unremarkable: WBC 5.0 x 10^3 cells/mm^3, BUN 7 mg/dL, and SCr 0.9 mg/dL. Today, his WBC is 24 x 10^3 cells/mm^3, BUN is 38 mg/dL, and SCr is 3.2 mg/dL, with 325 mL of UOP since his admission 24 hours ago.

1. Which best describes F.B.'s AKI?
   A. RIFLE class R.
   B. AKIN stage 1.
   C. RIFLE class F or AKIN stage 3.
   D. RIFLE class E or AKIN stage 3.

2. Which medication is most likely contributing to his AKI?
   A. Enalapril.
   B. Glipizide.
   C. Acetaminophen.
   D. Both enalaparil and glipizide equally.

3. When evaluating F.B.'s potential causes of AKI, which additional information or test would be most important to consider or obtain?
   A. Rate of loss, symptoms, and coexisting diseases and medications.
   B. Renal evaluation using ultrasonography because this can determine the cause of AKI in most patients.
   C. Review of blood chemistries because this will likely determine the cause of injury.
   D. Renal evaluation using biopsy.

II. KIDNEY REPLACEMENT THERAPIES

A. General Approaches to Managing AKI
   1. Once injury has occurred, therapy consists of providing supportive care and limiting additional insults, including nephrotoxins.
   2. In patients with shock, adequate fluid resuscitation should be initiated to restore effective circulation without producing volume overload.
   3. No specific pharmacologic therapy is effective in treating or reversing AKI.
   4. Metabolic and volume status should be followed closely, and KRT should be initiated when other approaches have failed.

5. Historically, KRT has been offered when a patient fulfills at least one of the AEIOUs: severe acidosis (A), electrolyte abnormalities (e.g. hyperkalemia) (E), intoxication (I), refractory volume overload (O), or symptomatic uremia (U). The multicenter BICAR-ICU study showed that randomization to sodium bicarbonate administration for a pH of 7.2 or less and PCO2 of 45 mm Hg or less, rather than no therapy, failed to improve all-cause mortality or death in ICU patients. However, of note, patients in the bicarbonate group had a significantly lower rate of new dialysis initiation, suggesting this temporizing measure has had a secondary benefit and dissuaded dialysis initiation for acidemia or hyperkalemia while kidneys spontaneously improved.
6. No consensus exists on the optimal timing of KRT. It is unclear whether KRT should be initiated when a patient reaches KDIGO stage III or earlier to avoid complications associated with metabolic/fluid derangements. Several recent randomized controlled trials regarding KRT timing have been published with disparate results. These include the ELAIN and AKIKI trials published in 2016, the IDEAL-ICU trial published in 2018, and the STARRT-AKI trial published in 2020.
    a. STARRT-AKI: Multicenter, multinational trial of 2927 individuals with stage 2 or 3 AKI who received either "accelerated KRT," meaning therapy was initiated within 12 hours of meeting eligibility criteria, or "standard KRT," meaning therapy was not initiated unless the patient fulfilled a conventional indication (analogous to the AEIOU indications mentioned earlier). Clinicians needed to attest to equipoise in the decision to initiate KRT before the patient could be enrolled. For the primary end point of all-cause mortality at 90 days, there was no difference between groups (44% in each group). Only two-thirds of the patients in the standard KRT group ultimately required dialysis, whereas 97% of the patients in the accelerated KRT group received dialysis. There were more adverse events in the accelerated KRT group.
    b. At this point, although the findings of these trials somewhat vary, it appears that there is no consistent, reproducible advantage to beginning KRT early in a patient's course as opposed to waiting until stage 3 AKI with an urgent indication.
7. The practice of KRT discontinuation is non-standardized but typically occurs when the precipitant is improved or resolved (e.g., sepsis controlled) and patients can satisfactorily maintain their fluid and solute balance without extracorporeal support.

B. Renal Replacement for AKI
    1. Modalities are classified according to duration as intermittent KRT, prolonged intermittent KRT, or continuous KRT.
    2. A defining feature of the type of KRT is the mechanism of solute and water transport through a semipermeable membrane by convection, diffusion, or both.
        a. Convection uses the concept of "solute drag" and can remove both small- (less than 500-1000 Da) and middle- (less than 10,000 Da) molecular-weight solutes. Solute removal occurs when the transmembrane pressure drives water and solute across a semipermeable membrane. This process involves adding replacement fluid (pre- and/or post-filter) to replace the excess volume that is being removed and replenish the desired electrolytes. Pre-filter replacement fluid may improve filter life and deter circuit clotting. Post-filter replacement fluid may improve solute clearance efficiency because of the hemoconcentration of blood entering the dialyzer.
        b. Diffusion is the movement of solutes from an area of higher solute concentration to an area of lower concentration. A concentration gradient is produced by running an electrolyte solution (i.e., dialysate fluid with a flow rate of 17–40 mL/minute) countercurrent to the flow of blood. Small-molecular-weight solutes (less than 500 Da) are cleared efficiently.

C. Intermittent Hemodialysis (IHD)
   1. IHD is the most common form of KRT used in the acute setting for treating patients with AKI or end-stage renal disease needing dialysis.
   2. Rapid metabolic clearance and/or fluid removal occurs over about 3- to 4-hour sessions.
      a. Intradialytic hypotension occurs in up to 30% of patients, which can lead to early dialysis termination or compromise of residual renal function. Several strategies to improve hemodynamics during IHD have been suggested, including fluid administration, temperature adjustment, ultrafiltration and sodium modeling, periprocedural antihypertensive hold, and midodrine and vasopressor therapy. Selection among these options is patient-specific and should only be done in conjunction with a careful evaluation of underlying etiologies for hypotension, including infection and cardiac function, among others.
      b. Patients with preexisting hypotension before IHD (e.g., septic shock), may not tolerate these large fluid and electrolyte shifts and may develop further deterioration of their hemodynamics.
      c. In patients with head trauma or hepatic encephalopathy, hypotension can reduce cerebral perfusion. Changes in solute concentrations can also worsen cerebral edema. Rapid plasma clearance of solutes acutely decreases their systemic intravascular concentrations while intracranial concentrations remain unchanged. This relatively high intracranial concentration of solute results in water transport back across the blood-brain barrier, which contributes to cell swelling, cerebral edema, and increased intracranial pressure.

D. Continuous KRT
   1. Continuous KRT is the most commonly used modality of KRT in hemodynamically unstable ICU patients. Past nomenclature of continuous KRT is continuous renal replacement therapy (CRRT) and may still be referred to as such in clinical practice or in historical literature.
   2. Continuous KRT modalities include continuous venovenous hemofiltration (CVVH), continuous venovenous hemodialysis (CVVHD), and continuous venovenous hemodiafiltration (CVVHDF).
   3. SCUF, or slow continuous ultrafiltration, is another type of continuous KRT that removes fluid without the need for replacement solutions. This therapy has limited effect on removal of waste products (e.g., BUN) or electrolytes and cannot correct acid-base abnormalities.
   4. Solute clearance during CVVHD occurs by diffusion.
   5. Solute clearance during CVVH occurs by convection, and the ultrafiltration rate determines the clearance rate for most solutes.
   6. For CVVHDF, solute removal occurs by both convection and diffusion, with diffusion commonly the dominant waste removal modality
   7. Because of the continuous nature of continuous KRT and prolonged exposure of blood to foreign surfaces of the extracorporeal circuit, anticoagulation is often required to prevent circuit clotting, maintain membrane permeability, and prevent blood loss in the clotted filter. Unfractionated heparin and regional citrate anticoagulation (RCA) are the most common anticoagulant options. Compared with systemic unfractionated heparin, RCA has been shown to increase filter life span and decrease bleeding, but with no difference in mortality. Challenges associated with RCA include accumulation in patients with reduced liver function and shock states, metabolic complications (acidosis, alkalosis, hypernatremia, hypocalcemia, and hypercalcemia), increased complexity, and need for use of a strict protocol. Systemic unfractionated heparin infusion remains an option for anticoagulation when unable to administer RCA. The recommended dose and monitoring is variable; however, it may be reasonable to titrate to a target activated partial thromboplastin time of 45–60 seconds or anti-Xa activity of 0.3–0.6 IU/mL. Heparin may also be administered regionally via infusion into the arterial line of the circuit with administration of continuous protamine post-filter to inactivate the unfractionated heparin.

E. Prolonged Intermittent KRT
   1. Includes sustained low-efficiency dialysis (SLED), sustained low-efficiency daily diafiltration (SLEDD-f), extended daily dialysis (EDD), and accelerated venovenous hemofiltration; also referred to as prolonged intermittent renal replacement therapy or PIRRT
   2. These therapies are a hybrid modality between intermittent and continuous options. Prolonged intermittent KRT is typically delivered using conventional hemodialysis machines with low blood-pump speeds (around 200 mL/minute) and low dialysate flow rates (around 300 mL/minute) for extended periods: 6–12 hours a day versus 3–4 hours for IHD or 24 hours for continuous KRT.
   3. Similar to continuous KRT, they allow for improved hemodynamic stability by producing gradual solute and volume removal compared with IHD.
   4. These therapies have certain advantages over continuous KRT. They produce high solute clearances, use existing IHD machines, eliminate the need for external solutions, and allow "time away" when various diagnostic and therapeutic procedures are needed, liberating health care workers from continuous involvement with the circuit. Disadvantages include limited data on drug clearance and a period in which solute and fluid management is paused (when off circuit).

F. Choosing a Mode
   1. Data are conflicting regarding the renal replacement mode of choice for critically ill patients.
   2. Outcomes such as mortality and renal recovery appear to be no different between IHD and continuous KRT; however, most studies are limited by design, patient characteristics, and crossover between different modalities.

G. Continuous KRT and Drug-Dosing Concepts
   1. Drug dosing during continuous KRT and SLED is often unclear because this information is not included in package inserts. Manufacturers are not required to study how these therapies alter clearance. General dosing considerations for continuous KRT include:
      a. For most medications, loading doses require no adjustment.
      b. If a drug is normally cleared by the kidneys or is removed by other KRT modalities, continuous KRT may significantly affect its removal.
      c. When available, drug-specific literature should be used to determine dose and frequency to minimize the likelihood of dosing errors. Although continuous KRT is meant to be continuous, it is often interrupted. If therapy is held for an extended period, dose adjustment may be required.
      d. Use caution when extrapolating historical literature to modern practice because current dialyzers can more readily clear small and middle MW molecules, and flow rates are higher than previously used. Collectively, these aspects would be expected to increase drug clearance and make historically recommended doses insufficient when applied to modern practice.
      e. With the many variables associated with continuous KRT delivery (e.g., modality, dialyzers, circuit downtime, patient instability), therapeutic drug monitoring should be used, when available.
   2. KRT modality, dialysis dose (e.g., more frequent IHD, higher dialysis dose/effluent flow rate increase drug elimination), and drug properties influence medication elimination. Effluent is any fluid leaving the circuit and not returning to the patient. This is spent fluid and could be dialysate or an ultrafiltrate from the blood.
   3. Drug proprieties that influence removal during continuous KRT include protein binding, molecular weight (MW), and volume of distribution (Vd). Drug charge is less important.
      a. The ability of a drug to bind to plasma protein (i.e., albumin) greatly influences how it is removed by continuous KRT. Removal is inversely proportional to the percent bound (i.e., the higher the percent bound, the less removed). Protein binding affects removal for both convection and diffusion.

b. Solutes are considered to have a small MW (less than 1000 Da), middle MW (1000-30,000 Da), or large MW (greater than 30,000 Da). Most unbound drugs have an MW less than 500 Da (e.g., cefepime 480 Da). Diffusive clearance (as with IHD and CVVHD) readily removes small MW drugs to a greater extent than middle MW drugs, whereas convective clearance modalities (as with CVVH) remove both small and middle MW drugs well. Highly protein bound drugs, as with albumin binding (64,000 Da), appear to have a high MW because of the association with the plasma protein. This large size would exceed the typical pore size of a high-flux, high-efficiency dialyzer, which is standard of care, and lead to reduced drug clearance.

c. Vd is inversely related to extracorporeal removal, wherein a larger Vd decreases elimination through the circuit because the drug is distributed throughout the body compartments. For continuous KRT, the slower rates allow for steady redistribution between body compartments, which can facilitate removal of drugs with a larger Vd to a greater degree than with IHD.

4. Continuous KRT modalities and their influence during therapy

a. CVVH

i. Solute removal during CVVH is by convection. Convection is influenced by the membrane pore size; the free fraction of drug, as discussed earlier; and the ultrafiltration rate.

ii. The ability of a substance to pass through a membrane by convection is termed the sieving coefficient (SC). The SC ranges from 0 to 1. A SC of 1 represents free movement, whereas a SC of 0 represents no movement across a filter.

iii. SC is best obtained from the primary literature or from patient-specific values. It can be calculated using a ratio of measured drug or other solute in the ultrafiltrate to its concentration in the plasma, $SC = C_{UF}/C_p$, where C_{UF} is concentration in the ultrafiltrate and C_p is concentration in the plasma.

iv. If a measured SC is not available, it can be estimated using the percent unbound to plasma protein, $SC = 1 - fb$, where fb is fraction bound (i.e., percent protein bound).

v. Site of replacement fluid administration can influence solute clearance. Adding replacement fluids pre-filter will dilute the blood entering the dialyzer. This diluted blood will have decreased drug concentrations; thus, less drug will be removed than if the replacement solution is infused after the filter.

If replacement fluids are administered postfilter (i.e., post-dilution), the clearance rate can be estimated using the following equation:

$$CVVH_{post\text{-}dilution} = Q_{UF} \times SC$$

If pre-dilution (i.e., before the filter) fluids are used, clearance across the membrane is reduced. Clearance can be estimated using the following equation:

$$CVVH_{pre\text{-}dilution} = Q_{UF} \times SC \times Q_b/(Q_b + Q_{rf})$$

where Q_{UF} is ultrafiltration flow rate, Q_b is blood flow rate, and Q_{rf} is pre-dilution replacement fluid flow rate. For pre-dilution fluid replacement to affect overall clearance, the rate must be high.

b. CVVHD

i. Solute removal during CVVHD occurs by passive diffusion. The flow of dialysate is countercurrent to that of the blood. Movement of solute across the semipermeable membrane occurs because of a concentration gradient, with movement from an area of higher concentration (blood) to an area of lower concentration (dialysate). This process occurs until equilibrium is established.

ii. Small substances (e.g., urea with an MW of 60 Da) are cleared more rapidly than larger substances (e.g., drugs with an MW greater than 500 Da).

iii. The ability of a drug to cross the dialysis filter during CVVHD is called the saturation coefficient (SA). Equations exist for estimating the SA when published data are unavailable. It can be calculated as $SA = C_E/C_p$, where C_E is the concentration in the effluent (spent dialysate) fluid and C_p is the concentration in plasma. The CVVHD clearance can then be approximated by Q_d x SA, where Q_d is the dialysate flow rate.

c. CVVHDF
   i. Solute removal during CVVHDF is by diffusion and convection (i.e., both dialysate and replacement fluids are used).
   ii. Clearances of small substances are about equal to the sum of the clearance from CVVH and CVVHD separately. However, as MW increases, this correlation no longer holds true.
   iii. Clearance is estimated as $CVVHDF = (Q_{UF} + Q_d) \times SA$.

d. Prolonged intermittent KRT
   i. Limited literature is available to guide drug dosing during prolonged intermittent KRT modalities such as SLED and EDD.
   ii. Extended dialytic therapies of 8-12 hours may alter drug dosing intervals. For example, drugs with frequent dosing intervals (e.g., time-dependent antibiotics like β-lactams) may need to be administered more often during the run, given by extended infusion, or repeated/readministered after the prolonged intermittent KRT treatment.
   iii. Solute removal during SLED/EDD is greater than that during CVVHD when estimated over the same time interval because higher dialysis flow rates are used during SLED/EDD treatments.
   iv. General dosing considerations for SLED
      (a) The duration of SLED and its flow rates (dialysate, blood) vary between studies and institutions, making a general approach to dosing problematic.
      (b) In addition, little information is available to guide drug dosing.
      (c) Like IHD and continuous KRT, the most important factors determining drug removal are protein binding, water solubility, MW (less than 500 kDa), and Vd (less than 0.8–1 L/kg); however, attention should also be given to the timing of drug administration relative to the prolonged intermittent KRT treatment.

Patient Case

Questions 4–6 pertain to the previous case.
F.B. is a 68-year-old man (weight 70 kg) admitted to your ICU with fever, elevated WBC, respiratory failure requiring mechanical ventilation, and norepinephrine to support his blood pressure. His medical history is significant for chronic back pain, diabetes, and hypertension. He takes both enalapril and glipizide once daily, as well as acetaminophen as needed. Before this admission, he had otherwise been healthy, seeing his primary care provider about 1 week ago. At that time, his blood pressure was 140/80 mm Hg, and his A1C was 5.2%. His laboratory workup was also unremarkable: WBC 5.0 x 10^3 cells/mm^3, BUN 7 mg/dL, and SCr 0.9 mg/dL. Today, his WBC is 24 x 10^3 cells/mm^3, BUN 38 mg/dL, and SCr 3.2 mg/dL, with 325 mL of UOP since his admission 24 hours ago. It is determined that F.B. needs KRT to manage his volume and control his metabolic derangements. He is currently receiving norepinephrine with a mean arterial pressure of 65 mm Hg.

4. Which renal replacement mode will most likely be chosen?
   A. IHD.
   B. Slow-low EDD.
   C. Continuous KRT.
   D. Peritoneal dialysis.

5. F.B. will be initiated on a new extended-spectrum cephalosporin with limited information available for its use during continuous KRT. Given the following parameters, estimate the drug's SC: MW 480 Da, positively charged drug, Vd 1.9 L/kg, protein binding 12%.
   A. 0
   B. 0.52
   C. 1.9
   D. 0.88

6. Given the calculated SC, estimate the new cephalosporin's clearance during continuous KRT if the prescription is CVVH, blood flow 200 mL/minute, ultrafiltration rate 2000 mL/hour, 100% pre-filter replacement fluids, and use of a high-flux, high-efficiency dialyzer.
   A. 0.9 L/hour.
   B. 1.5 L/hour.
   C. 3.3 L/hour.
   D. 5.3 L/hour.

REFERENCES

Acute Kidney Injury

1. Barlam TF, Cosgrove SE, Abbo LM, et al. Implementing an antibiotic stewardship program: guidelines by the Infectious Diseases Society of America and the Society for Healthcare Epidemiology of America. Clin Infect Dis 2016;62:e51-77.
2. Bellomo R, Ronco C, Kellum JA, et al. Acute Dialysis Quality Initiative Workgroup. Acute renal failure—definition, outcome measures, animal models, fluid therapy and information technology needs: the Second International Consensus Conference of the Acute Dialysis Quality Initiative (ADQI) Group. Crit Care 2004;8:R204-12. Original report describing the RIFLE criteria and providing a rationale for their conception.
3. Bentley ML, Corwin HL, Dasta J. Drug-induced acute kidney injury in the critically ill adult: recognition and prevention strategies. Crit Care Med 2010;38:S169-74. Provides a good overview of the mechanisms of drug-induced kidney injury, recognition of AKI, and AKI prevention.
4. Chen S. Retooling the creatinine clearance equation to estimate kinetic GFR when the plasma creatinine is changing acutely. J Am Soc Nephrol 2013;24:877-88.
5. Göcze I, Jauch D, Götz M, et al. Biomarker-guided intervention to prevent acute kidney injury after major surgery: the prospective randomized BigpAK study. Ann Surg 2018;267:1013-20.
6. Kane-Gill SL. Nephrotoxin stewardship. Crit Care Clin 2021;37:303-20.
7. Kellum JA, Romagnani P, Ashuntantang G, et al. Acute kidney injury. Nat Rev Dis Primers 2021;7:1-17.
8. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney Int Suppl 2012;2:1-138. Updated clinical practice guideline for AKI.
9. McCullough PA, Bouchard J, Waikar SS, et al. Implementation of novel biomarkers in the diagnosis, prognosis, and management of acute kidney injury: executive summary from the tenth consensus conference of the Acute Dialysis Quality Initiative (ADQI). Contrib Nephrol. 2013;182:5-12. Summarizes the role for novel biomarkers in AKI assessment.
10. Mehta RL, Awdishu L, Davenport A, et al. Phenotype standardization for drug-induced kidney disease. Kidney Int 2015;88:226-34.
11. Karimazadeh I, Barreto EF, Kellum JA, et al. Moving toward a contemporary classification of drug-induced kidney disease. Crit Care 2023;47:435.
12. Miano TA, Hennessy S, Yang W, et al. Association of vancomycin plus piperacillin-tazobactam with early changes in creatinine versus cystatin C in critically ill adults: a prospective cohort study. Intensive Care Med 2022;48:1144-55.
13. Ostermann M, Bellomo R, Burdmann EA, et al. Controversies in acute kidney injury: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Conference. Kidney Int 2020;98:294-309.
14. O'Sullivan ED, Doyle A. The clinical utility of kinetic glomerular filtration rate. Clin Kidney J 2017;10:202-8.
15. Pais GM, Liu J, Avedissian SN, et al. Lack of synergistic nephrotoxicity between vancomycin and piperacillin/tazobactam in a rat model and a confirmatory cellular model. J Antimicrob Chemother 2020;75:1228-36.
16. Schreier DJ, Kashani KB, Sakhuja A, et al. Incidence of acute kidney injury among critically ill patients with brief empiric use of anti-pseudomonal beta-lactams with vancomycin. Clin Infect Dis 2018. [Epub ahead of print]
17. Silver S, Adhikari NK, Chan C, et al. Nephrologist follow-up versus usual care after an acute kidney injury hospitalization (FUSION). Clin J Am Soc Nephrol 2021;16:1005-14.
18. Uchino S, Kellum JA, Bellomo R, et al. Acute renal failure in critically ill patients: a multinational, multicenter study. JAMA 2005;294:813-8. A prospective, observational, multinational study of ICU patients to determine the prevalence of acute renal failure; characterize its etiology, illness severity, and clinical practice; and investigate the impact of these differences on patient outcomes.
19. Zarbock A, Kullmar M, Ostermann M, et al. Prevention of cardiac surgery–associated acute kidney injury by implementing the KDIGO guidelines in high-risk patients identified by biomarkers: the PrevAKI-multicenter randomized controlled trial. Anesth Analg 2021;133:292-302

Kidney Replacement Therapies

1. Bagshaw SM, Wald R, Adhikari N, et al. Timing of initiation of renal-replacement therapy in acute kidney injury. N Engl J Med 2020;383:40-51.
2. Bogard KN, Peterson NT, Plumb TJ, et al. Antibiotic dosing during sustained low-efficiency dialysis: special considerations in adult critically ill patients. Crit Care Med 2011;39:560-70. Addresses issues of antibiotic dosing during SLED and provides an experience with it.
3. Gaudry S, Palevsky PM, Dreyfuss D. Extracorporeal kidney-replacement therapy for acute kidney injury. N Engl J Med 2022;386:964-75.
4. Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. Pharmacotherapy 2009;29:562-77. Provides a critical review of current literature related to the influence of CKD and AKI on the pharmacokinetic and pharmacodynamic properties of antimicrobial agents and suggests dosing strategies for select antimicrobial agents.
5. Jaber S, Paugam C, Futier E, et al. Sodium bicarbonate therapy for patients with severe metabolic acidemia in the intensive care unit (BICAR-ICU): a multicenter, open-label, randomized controlled, phase 3 trial. Lancet 2018;392:31-40.
6. Maker JH, Dager WE, et al. Antibiotic dosing for critically ill adult patients receiving intermittent hemodialysis, prolonged intermittent renal replacement therapy, and continuous renal replacement therapy: an update. Ann Pharmacother 2020;54:45-55.
7. Tolwani A. Continuous renal-replacement therapy for acute kidney injury. N Engl J Med 2012;367:2505-14. A good overview of solute clearance during CRRT; lists indications and contraindications and clinical use.
8. Zarbock A, Kullmar M, Kindgen-Milles D, et al. Effect of regional citrate anticoagulation vs systemic heparin anticoagulation during continuous kidney replacement therapy on dialysis filter life span and mortality among critically ill patients with acute kidney injury. JAMA 2020;324:1629-39.

ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: C**

Both RIFLE class "F" and AKIN stage 3 are met using similar SCr and UOP criteria. For this case, the SCr increased by at least 3-fold above baseline, and the patient's UOP was less than 0.3 mL/kg/hour for 24 hours. Although A and B are true, the worse values in the RIFLE class and AKIN staging systems should be chosen when determining the class and stage of AKI. Answer D is incorrect because his renal indices have not been present for at least 3 months. Collectively, both the RIFLE and the AKIN criteria have been included in the KDIGO criteria, which is the current standard definition.

2. **Answer: A**

This patient has severe sepsis that led to decreased renal perfusion, causing AKI. Enalapril likely contributed to the injury by altering renal hemodynamics (Answer A is correct). Neither glipizide nor acetaminophen is likely to cause AKI (Answers B–D are incorrect).

3. **Answer: A**

Acute kidney injury cannot currently be diagnosed with a specific blood test or imaging study (Answer C is incorrect). The degree of kidney dysfunction and the patient's symptoms and coexisting diseases may provide clues regarding the etiology of the patient's kidney injury (Answer A is correct). Drug-induced AKI is most common in the setting of additional insults and may occur even after the drug is discontinued. Given a patient's history and presentation as well as a patient's physical examination, a clinician may pursue additional testing to find the DIKD phenotype. Serum markers (e.g., SCr, UOP) are often nonspecific (Answer C is incorrect); kidney biopsy is only warranted for certain DIKD phenotypes (e.g., AIN, glomerular diseases) and is often not pursued in the acute setting (Answer D is incorrect). Renal ultrasonography would be helpful for nephrolithiasis/crystalluria phenotypes or evidence of postrenal AKI not alleviated with urinary catheter placement (Answer B is incorrect).

4. **Answer: C**

Intermittent hemodialysis is often used in critically ill patients because many physicians are familiar with this therapy; however, about 20%–30% of patients receiving IHD become hypotensive and require discontinuation or a switch to an alternative therapy. Although prolonged intermittent KRT methods like SLED or EDD are an option, some form of continuous KRT would likely be chosen because of this patient's unfavorable hemodynamics. Continuous renal replacement therapies such as CVVH, CVVHD, and CVVHDF are often used because they allow for slower flow rates and improved hemodynamics. However, there is no clear benefit with one therapy over another.

5. **Answer: D**

Protein binding primarily determines drug clearance during KRT. Other factors that contribute to drug clearance include low MW and low Vd. Drug charge leading to dialyzer adhesion has only a limited effect on drug clearance during continuous KRT and has been poorly studied. When available, SC should be identified from the package insert information or the primary literature or calculated from the concentration of the solute in the ultrafiltrate divided by the concentration of the solute in the plasma. When unavailable, SC can be estimated as SC = 1 − fb. In this case, the SC = 1 − 0.12 = ~0.88 (Answer D is correct; Answers A–C are incorrect).

6. **Answer: B**

The general approach to calculating clearance during CVVH multiplies the SC (the degree to which a substance is able to pass through the membrane) by the ultrafiltration rate. If predilution (i.e., before the filter) fluids are used, clearance across the membrane is reduced, which must be accounted for in the formula. Clearance in that case can be estimated using the following equation: CVVHpre-dilution = Q^{UF} x SC x [Q^b/(Q^b + Q^{rf})] (where Q^{UF} is ultrafiltration flow rate (liters per hour), Q^b is blood flow rate [converted to liters per hour for consistent units], and Q^{rf} is predilution replacement fluid flow rate, as stated earlier). In this case, clearance = 2 L/hr x 0.88 x [12 L/hr/(12 L/hr + 2 L/hr)] = 1.5 L/hr (Answer B is correct; Answers A, C, and D are incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: A**

The drug-induced kidney disease phenotype most likely associated with these antibiotics is AIN from ceftriaxone. Acute interstitial nephritis is considered under the AKI phenotype (Answer A is correct). Azithromycin would unlikely cause drug-induced kidney disease. Although AIN is an overall uncommon cause of DIKD, in 75% of AIN cases, one or more drugs are involved. Several medications have been associated with this type of kidney disease, including penicillins, cephalosporins, sulfonamides, ciprofloxacin, vancomycin, NSAIDs and COX-2 inhibitors, proton pump inhibitors, immune checkpoint inhibitors, antiepileptic drugs, ranitidine, diuretics, and cocaine. This patient may have other contributing factors to AKI, including sepsis and shock. Glomerular disease, nephrolithiasis, and tubular dysfunction have not been associated with cephalosporins (Answers B-D are incorrect).

2. **Answer: C**

Stage 3 AKI is met by both the SCr criteria (3 x baseline) and the need for KRT (Answer C is correct; Answers A and B are incorrect). Urinary output is not needed to stage the kidney dysfunction in this case because the patient fulfills other criteria (Answer D is incorrect).

3. **Answer: C**

Answer C is correct. Drug-dosing recommendations for IHD can be found in many resources. However, dosing during continuous KRT and SLED/EDD is less clear. Primary literature and/or summary tables for continuous KRT and SLED should be referenced because these recommendations are not usually found in other sources (Answers A and B are incorrect). Use caution to ensure that identical modes of continuous KRT are referenced with similar flow rates. Drug dosing using a sieving coefficient should only be used when a review of the literature fails to provide specific drug adjustment recommendations (Answer D is incorrect).

4. **Answer: A**

Solute removal during CVVH is by convection, which is primarily influenced by membrane pore size, free fraction of drug, and ultrafiltration rate. Diffusion is the process of clearance during CVVHD making Answers B and C incorrect. Membrane binding does occur but is not the primary route of clearance during CVVH (Answer D is incorrect).

5. **Answer: A**

Answer A is correct. Protein binding is a primary determinant of drug clearance during continuous KRT. Only unbound drug can be cleared through the circuit; therefore, an increase in protein binding while other factors remain stable would lead to reduced drug elimination (Answers B and C are incorrect). A high Vd would reduce drug clearance (Answer D is incorrect). The impact of high Vd is more significant in IHD, resulting in less drug removal because there is little time during the 3- to 4-hour IHD run for a drug to redistribute out of the peripheral tissues.